From Cultural Knowledge to Epistemic Behavior: Social-Epistemic Stress Tests for Culturally Grounded LLM Evaluation

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Abstract

Evaluations of cultural competence in large language models (LLMs) increasingly assess cultural knowledge, geographic sensitivity, and contextual information in downstream interactions. These approaches show that cultural knowledge is unevenly represented and does not reliably translate into appropriate behavior. A related problem remains under-examined: how do interventions intended to make an LLM more context-sensitive alter what it recognizes, infers, and presents as authoritative? We advance the position that culturally competent LLM behavior should be evaluated as warranted context sensitivity rather than cultural knowledge alone. We introduce a social-epistemic stress-testing framework that treats clinical guardrails, situated knowledge, and retrieval-augmented evidence as experimental variables rather than presumed improvements. The framework spans five epistemic configurations, with perturbations to labels, distress expression, geographic cues, social context, and evidence scope. We describe a provenance-preserving process that translates qualitative knowledge into auditable model instructions without converting situated observations into cultural rules. We hypothesize that interventions intended to increase context sensitivity may correct some failures while introducing others, including contextual over-inference, evidence-scope violations, or authoritative unsupported inference. We will empirically evaluate this hypothesis using perinatal mental health scenarios developed with experts across Ghana, Nigeria, Ethiopia, and Rwanda, country contexts selected pragmatically based on existing collaborations and relevant evidence. This work contributes a method for evaluating culturally grounded LLM behavior wherever socially situated experience and heterogeneous evidence shape model decisions.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; • **Human-centered computing** → **HCI theory, concepts and models**; • **Social and professional topics** → *Cultural characteristics.*

Keywords

Cultural competence, Large language models, Sociotechnical evaluation, Situated knowledge

1 Introduction

Determining whether an LLM is culturally competent is fundamentally a measurement problem [4, 19]. A central question is therefore: *what behaviors should count as evidence of cultural competence, and which should indicate failure?* Because constructs such as fairness, risk, and cultural competence are not directly observable, these measurement choices can introduce assumptions that diverge from the intended evaluation target [17]. This concern is acute for generative systems because responses may appear factually plausible and socially appropriate while relying on unsupported assumptions about people or settings. Machine-learning evaluations often privilege decontextualized performance even when deployment decisions depend on social context, responsibility, and harm [16]. Evaluating cultural competence therefore requires specifying what competence means, which failures matter, and whose knowledge informs those judgments [8].

Recent work has progressively expanded cultural-competence evaluation beyond factual recall. CulturalBench measures cultural knowledge across 45 global regions and documents geographic performance gaps [9], while CultureBank preserves multiple perspectives rather than treating cultural knowledge as a single canonical account [28]. Other approaches move further toward evaluating contextual use: Bhatt and Diaz (2024) examine open-ended responses under nationality perturbations, SocialCC tests whether cultural knowledge is appropriately applied during multi-turn interactions, and Saha et al. (2025) distinguish meta-cultural competence from static cultural knowledge [5, 6, 24, 32]. Together, this work shows that cultural competence cannot be reduced to whether a model possesses cultural facts; it also depends on whether the model uses contextual information appropriately.

This progression motivates a further question that existing evaluations address less directly: *how does model behavior change when different forms of contextual knowledge are deliberately introduced into the system?* That question is increasingly important because contemporary LLM applications rarely rely on parametric knowledge alone. They may also incorporate developer instructions, safety rules, retrieved documents, and knowledge elicited from affected people, each of which changes the information available to the model and the conditions under which it reasons. Although these mechanisms are often introduced as mitigations, their effects are not necessarily uniformly beneficial. Lertvittayakumjorn et al. (2025), for example, found that search grounding improved propositional cultural knowledge while increasing stereotypical judgments, while cross-cultural retrieval can also import biases from the retrieved collection [20, 23]. The resulting evaluation problem is therefore not simply whether additional knowledge improves performance, but how different knowledge interventions change what a model recognizes, what it infers from contextual cues, and which claims it presents with greater certainty.

We operationalize culturally competent LLM behavior as the capacity to use socially and culturally relevant information when supported by the interaction or evidence, while avoiding unsupported inference from group membership and preserving the provenance,

scope, and uncertainty of contextual knowledge. We therefore argue that cultural competence should not be evaluated primarily by how much cultural knowledge a model possesses. Evaluation should test whether contextual knowledge changes behavior when evidentially relevant, whether the model remains restrained when group membership is not determinative, and whether knowledge introduced through guardrails and retrieval retains its provenance and uncertainty. Building on this position, we propose social-epistemic stress testing: varying the knowledge available during inference and crossing those changes with controlled perturbations of socially meaningful cues.

Our central hypothesis is that *interventions intended to increase context sensitivity will have non-monotonic effects: clinical guardrails, situated knowledge, and retrieval may correct particular failures while introducing contextual over-inference, evidence-scope violations, or more authoritative unsupported inference.* Our contribution is a framework that makes the mitigation itself an object of evaluation rather than assuming that additional knowledge necessarily improves cultural competence.

We use three evaluation questions to make this position empirically testable:

EQ1: How does model behavior change when formal clinical knowledge, situated knowledge, and retrieved scientific evidence are introduced under otherwise equivalent task conditions?

EQ2: Do models remain stable under non-determinative contextual perturbations while responding appropriately to socially consequential information?

EQ3: Do knowledge interventions preserve evidence provenance and uncertainty, or do they introduce contextual over-inference, evidence-scope violations, or other forms of epistemic overreach?

We develop the framework through perinatal mental health, where clinical meaning, social experience, and evidence provenance intersect. Although the empirical study tests the position advanced above, the evaluation problem extends to other domains where socially situated experience and bounded evidence shape model behavior. Testing this position requires revealing not only whether an answer is acceptable, but which information and knowledge sources it depends on.

2 From Robustness Testing to Epistemic Stress Testing

Controlled perturbation provides one way to make those dependencies visible, as recent health-AI evaluations illustrate. For instance, Gu et al. (2026) show that strong health-AI benchmark scores can conceal fragility: frontier models answered diagnostic questions after necessary visual information was removed, changed behavior under answer-order or distractor perturbations, and sometimes produced fluent explanations containing unsupported evidence [13]. Their findings shift evaluation from whether a model reaches the expected answer to whether that answer remains grounded in the information that should support it. We use epistemic behavior to refer to observable output changes when model knowledge or contextual evidence is systematically altered.

Yu et al. (2026) identify a related problem when geographic knowledge enters maternal and neonatal health reasoning [33]. Across 204 countries, GPT-4o generated cases that did not reflect observed prevalence, supplied pregnancy-related statistics without acknowledging unavailable regional evidence in 44% of relevant cases, and produced erroneous citations in 46.7%. Some examination errors occurred when regional prevalence appeared to outweigh case symptoms. Geographic knowledge can therefore be correct yet used inappropriately as a case-level prior or support for a broader population claim.

Together, these studies motivate a shift in the object of evaluation. We make the model-side knowledge arrangement an experimental variable and treat guardrails, situated knowledge, and retrieval as interventions to audit rather than improvements to assume. This follows the sociotechnical principle that systems cannot be evaluated apart from their abstractions and assumptions and responds to calls to examine which values and experiences evaluations center [3, 8, 10, 25, 27]. Making these assumptions experimentally visible requires a framework that varies both the knowledge available to the model and the contextual evidence presented to it.

3 Social-Epistemic Evaluation Framework

Our social-epistemic evaluation framework operationalizes this approach by crossing five epistemic configurations with five controlled stress tests. The first axis addresses EQ1 by varying the forms of knowledge available during inference. The second addresses EQ2 and EQ3 by perturbing clinical, social, geographic, and evidentiary cues while preserving a prespecified core.

3.1 Five Epistemic Configurations

We define an epistemic configuration as the researcher-controlled arrangement of parametric, instructional, situated, and retrieved knowledge available during inference (see Table 1). These configurations are not ordered from least to most culturally competent. Clinical instructions may improve diagnostic boundaries while privileging particular symptom expressions (i.e., C2a); situated knowledge may improve recognition without biomedical terminology while increasing the risk that contextual cues become cultural templates (i.e., C2b); and retrieval may provide relevant evidence without preserving its limits (i.e., C3a and C3b). Because retrieval changes both what the model can access and what appears authoritative at inference time, we evaluate whether retrieved evidence remains bounded by its population, setting, measurement, and methodological scope [21, 22].

Comparisons across configurations identify which intervention is associated with a behavioral change. C2a may improve escalation without improving recognition of indirectly expressed distress; C2b may alter recognition; and C3a may improve citation fidelity while still generalizing facility-based evidence to a national population. These contrasts show which failures an intervention addresses and which new errors it may introduce.

3.2 Controlled Social-Epistemic Stress Tests

The second axis changes one stimulus feature while preserving the clinically or evidentially relevant core. Each test has a prespecified invariant: model behavior should remain stable unless the perturbation is legitimately relevant.

Table 1: Five epistemic configurations and their evaluation purposes.

Configuration	Knowledge available	Evaluation purpose
C1: Baseline	Provider-aligned parametric knowledge	Establish unmodified behavior
C2a: Clinical guardrails	C1 + formal clinical and safety constraints	Test explicit clinical boundaries
C2b: Situated-knowledge-informed guardrails	C2a + instructions derived from patient and clinician situated knowledge	Test effects on recognition and contextual interpretation
C3a: Clinical + retrieval	C2a + provenance-preserving scientific retrieval	Test effects of external evidence
C3b: Situated knowledge + retrieval	C2b + the same retrieval architecture	Test interactions between situated and scientific knowledge

ST1: Clinical-label removal preserves symptoms, duration, impairment, and safety information while removing explicit labels such as "depression."

ST2: Distress-expression perturbation presents clinically equivalent information through direct and indirect descriptions. Together, ST1 and ST2 test *recognition*: whether the model identifies the same underlying clinical pattern without relying on explicit psychiatric terminology.

ST3: Country-cue substitution changes only the country identifier while preserving case facts. Because country is not treated as culture, ST3 tests *unsupported group inference*: whether geographic identity alone alters symptom recognition, severity assessment, or diagnostic certainty.

ST4: Social-context perturbation varies consequential information such as stigma, family opposition, or access to support while holding the clinical pattern constant. ST4 tests *appropriate contextual use*: whether the model responds to explicitly provided social information without generalizing it to others who share a country or group identity.

ST5: Evidence-scope stress testing varies the scope and type of evidence, including national, regional, facility-based, validation, screening, and qualitative sources. It tests *provenance, scope, and uncertainty*: whether the model preserves the population, setting, measurement, and methodological limits of the evidence rather than expanding bounded claims beyond what the source supports.

Across these tests, we evaluate four core failure modes. Epistemic misrecognition occurs when consequential information becomes less legible because it falls outside a privileged vocabulary or form. Contextual over-inference occurs when a model infers beliefs, risks, preferences, or explanations beyond what contextual cues support. Evidence-scope violation occurs when a claim exceeds the population, setting, measurement, or methodology of its source. Fluent epistemic overreach occurs when limited or uncertain evidence is presented with unwarranted authority. Health-specific review also examines diagnostic-boundary violations and safety miscalibration. These constructs complement accuracy by examining the legitimacy of evidence and assumptions supporting a response. Some of these failures can arise from the very contextual knowledge introduced to improve model behavior. Therefore, the framework must also specify how situated knowledge is translated into model interventions without hardening bounded observations into generalized cultural assumptions.

3.3 Translating Situated Knowledge into Auditable Artifacts

That translation introduces a further methodological problem: how should locally situated knowledge enter a model without becoming a fixed rule about a population? Feminist epistemology argues that knowledge is situated and that social position shapes what can be observed and interpreted, while no standpoint provides a complete account [11, 15, 18, 29]. For AI evaluation, this creates two obligations: 1) epistemic inclusion, such that knowledge produced by people affected by a system informs what the benchmark treats as consequential; and 2) epistemic restraint, such that findings from bounded qualitative samples retain their contextual limits rather than becoming generalized cultural rules. Together, these obligations define a central tension for culturally grounded evaluation: situated knowledge must meaningfully inform the benchmark without becoming generalized claims about a population.

We operationalize this tension through a provenance-preserving knowledge-to-artifact chain:

$$\textit{qualitative finding} \rightarrow \textit{epistemic concern} \rightarrow \textit{anticipated model failure} \rightarrow \textit{guardrail principle} \rightarrow \textit{exact model instruction} \rightarrow \textit{stress-test stimulus} \rightarrow \textit{evaluation criterion}$$

The knowledge-to-artifact chain makes the movement from situated evidence to model intervention explicit: a qualitative finding identifies a condition under which a model may fail, which is then translated into a guardrail principle and controlled test. This reduces the risk of converting a bounded qualitative observation into a cultural generalization.

Consider recognition of perinatal distress when a person does not use explicit psychiatric terminology. Qualitative studies in African settings document descriptions including "thinking too much," headache, weakness, interpersonal strain, spiritual concerns, and other embodied or socially situated expressions [1, 7, 12, 26]. These findings do not establish a uniform way that women in Ghana, Ethiopia, or Nigeria express depression, nor do they make any expression a psychiatric symptom independent of clinical context. Instead, they identify a testable model dependency: an LLM that associates clinically consequential distress primarily with explicit psychiatric labels may respond differently to equivalent symptom information when it is described indirectly.

The framework converts that concern into a testable intervention rather than a cultural rule. The guardrail principle is that recognition should depend on case information—including symptom pattern, persistence, functional impact, and safety indicators—rather than psychiatric self-identification. The derived model instruction directs attention to this evidence without assigning predetermined clinical meaning to a cultural expression. ST1 then holds clinically relevant features constant while removing the explicit psychiatric

label and tests whether recognition, assessment of concern, and recommendations for human follow-up remain appropriately calibrated.

Situated knowledge therefore identifies a model dependency to investigate; it does not determine the correct interpretation of a person based on cultural membership. Because the artifact remains traceable to its source evidence, reviewers can assess whether the intervention improves recognition or introduces diagnostic overreach or contextual over-inference.

The knowledge-to-artifact process thus creates an auditable path from situated evidence to an experimentally testable dependency while preserving the partial and bounded character of the source knowledge. These principles guide the empirical study described next, where socially situated experience and the scope of available evidence make the distinction between contextual sensitivity and unsupported generalization consequential.

4 Planned Instantiation in Perinatal Mental Health

The empirical study described here tests our position on culturally grounded model evaluation rather than assuming its validity. Perinatal mental health is useful because both lived experience and its measurement are context-dependent. Tsai et al. (2013) reviewed 25 studies involving 12,544 women across nine African countries and found that only three used instruments developed for a specific cultural setting [30]. Gyimah et al. (2024) identified 47 maternal mental-health screening tools in sub-Saharan African primary-care studies and reported substantial methodological variation and incomplete psychometric reporting [14]. Together, these findings show that prevalence estimates depend on population, timing, instrument, threshold, language, and validation context; removing these attributes can turn methodological heterogeneity into apparent population difference.

The study focuses on Ghana, Nigeria, Ethiopia, and Rwanda as a convenience sample of country contexts selected on the basis of existing collaborations and available perinatal mental-health evidence. Their value is analytic contrast rather than representation of Africa or treatment as homogeneous cultures. Existing studies document variation in distress expression, explanatory models, disclosure, help-seeking, social relationships, and service barriers [1, 2, 7, 26, 31]. We treat these findings as evidence about specific participants and settings rather than national attributes. This distinction is particularly important when evaluating models outside dominant Western benchmark contexts, where clinical constructs, benchmark conventions, and safety assumptions may reflect evidence and interaction norms concentrated in Western settings. The design therefore tests whether models respond appropriately to locally situated information that is actually present while avoiding assumptions triggered by geographic identity alone.

Within this design, country serves three analytic functions: an evidence-provenance boundary, a context for empirically grounded stimulus construction, and a counterfactual cue whose independent effect can be tested. A stated disclosure barrier should influence a response regardless of country, whereas changing only the country label should not introduce assumptions about religion, family structure, adherence, resilience, or help-seeking. The initial English-only benchmark therefore evaluates contextual and evidentiary dependencies rather than multilingual competence. The benchmark will be newly constructed rather than adapted from an existing dataset and will use synthetic, fictional scenarios grounded in published clinical, qualitative, epidemiologic, and measurement evidence from the four study settings rather than patient-level data. The evaluation will use two stimulus banks: a clinical-context bank of synthetic vignettes spanning normative adjustment, depressive symptoms, functional impairment, embodied distress, bonding difficulty, constrained disclosure, suicidal ideation, and possible severe psychiatric risk; and an evidence-reasoning bank testing prevalence, screening tools, thresholds, validation, citations, and evidence availability. Matched vignette families will manipulate one intended feature at a time. Before model evaluation, multidisciplinary reviewers will independently assess and annotate scenarios for clinical equivalence, contextual plausibility, evidentiary scope, and preservation of the intended stress-test invariant, revising or excluding scenarios when these criteria are not met. Retrieval conditions will use a curated corpus retaining metadata for country, setting, sample, measurement instrument, threshold, and relevant validation limitations. Vignettes, reviewer labels, expected answers, and prior outputs will remain outside the retrieval corpus. The initial evaluation will compare GPT-5.6 Sol and Gemini 3.8 Flash through their respective provider APIs; we will record exact model identifiers and versions, access dates, and supported inference settings, and hold prompts and inference parameters constant within each model across epistemic configurations except for the experimentally manipulated knowledge intervention.

Human review will be distributed by expertise: perinatal mental-health clinicians will assess recognition, diagnostic boundaries, and escalation; epidemiologic or research-methods reviewers will assess evidence scope, measurement, and uncertainty; and country-context and lived-experience reviewers will assess unsupported inference, stigmatizing assumptions, and distortions of social meaning. Reviewers will score outputs independently and without model or configuration labels where feasible. We describe the framework as situated-knowledge-informed rather than treating published qualitative evidence as participatory design, while existing collaborations support locally informed contextual review. For reproducibility, we will document model-facing prompts, guardrail derivations, stress-test invariants, provenance metadata, model/version identifiers, inference settings, and reviewer criteria.

5 Conclusion

We argue that culturally grounded LLM evaluation should assess not only what information a model receives, but how different knowledge configurations shape what it recognizes, infers, and presents as authoritative. Social-epistemic stress testing operationalizes this position by treating clinical guardrails, situated knowledge, and retrieved evidence as interventions to audit through controlled contextual and evidentiary perturbations. The planned study tests whether these interventions produce warranted context sensitivity or instead shift failure into new forms. The same evaluation logic can extend to other domains in which socially situated experience and bounded evidence jointly shape model behavior.

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